

Netflix Movies & TV Shows Dataset Analysis Report

Prepared using Microsoft SQL Server.

Task 1: Dataset Understanding

The Netflix Movies & TV Shows dataset contains information about movies and TV shows available on Netflix, including the title, content type, director, cast, country, date added, release year, rating, duration, genres and description.

Rows: 8,807

Columns: 12

Numerical feature: release_year

Categorical features: show_id, type, title, director, cast, country, date_added, rating, duration, listed_in, description

Primary key candidate: show_id

Task 2: Data Cleaning

Missing values in director, cast and country were replaced with 'Unknown'. Missing ratings were replaced with 'Not Rated'. Three misplaced duration values were corrected by moving them from the rating column to the duration column. Missing date_added values were left as NULL because the correct values could not be determined. No duplicate records were found. Date formats, text formatting, column names and data types were reviewed and found to be consistent. Validation confirmed no invalid values, inconsistent records, incorrect formats or unrealistic release years.

Cleaning Summary

Issue Found	Action Taken
Missing Values	Filled or corrected where appropriate; retained NULL date_added values.
Duplicates	None found.
Invalid Entries	None found.
Standardization	Verified and applied where necessary.

Task 3: Exploratory Data Analysis

Summary Statistics (release_year): Mean = 2014.18, Median = 2017, Minimum = 1925, Maximum = 2021, Standard Deviation = 8.82.

Movies vs TV Shows

Movies: 6,131; TV Shows: 2,676. Movies make up most of the catalogue.

Content Added by Year

Content additions peaked in 2019 with 1,999 titles.

Top Content-Producing Countries

United States ranked first, followed by India and the United Kingdom.

Most Common Ratings

TV-MA was the most common rating, followed by TV-14.

Most Common Genres

Dramas and International Movies were the most common genre combination.

Section 4: Visualizations

Visualization 1: Movies vs TV Shows Distribution

Purpose: This chart compares the number of Movies and TV Shows available on Netflix.

Finding: Movies make up the larger share of the catalog, showing that Netflix offers more movies than TV shows.

Visualization 2: Content Added by Year

Purpose: Shows how many titles were added to Netflix each year.

Finding: Content additions increased steadily and peaked around 2019 before declining slightly in later years.

Visualization 3: Top 10 Content-Producing Countries

Purpose: Shows the countries with the highest number of Netflix titles.

Finding: The United States contributes the most titles, followed by India and the United Kingdom.

Visualization 4: Most Common Content Ratings

Purpose: Displays the most frequently used content ratings.

Finding: TV-MA and TV-14 are the most common ratings, suggesting much of the library targets mature and teenage audiences.

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Visualization 5: Top 10 Most Common Genres

Purpose: Shows the genres that appear most often in the dataset.

Finding: Drama-related categories and documentaries appear more frequently than other genres.

Task 5: Key Insights

1. Movies account for a much larger share of the Netflix library than TV Shows, indicating that films remain the platform's primary content type.
2. Netflix expanded its catalog rapidly between 2016 and 2019, with 2019 recording the highest number of titles added.
3. The United States is the leading content-producing country in the dataset, while India and the United Kingdom also make significant contributions.
4. Most titles are rated TV-MA or TV-14, showing that a large portion of Netflix's content is aimed at older audiences.
5. Drama-related genres appear more often than any other category, highlighting the popularity of drama content on the platform.

Cleaning Challenges Encountered

- Missing values were found in director, cast, country, rating and date_added.
- Missing text values were replaced with appropriate placeholders.
- Some duration values were corrected after being found in the rating column.
- Missing date_added values were left as NULL because the correct dates could not be verified.
- No duplicate records were found.

Key EDA Findings

Movies outnumber TV Shows.

Content additions peaked in 2019.

The United States has the highest number of titles.

TV-MA is the most common rating.

Drama-related genres are the most common.

Top Insights

Netflix's catalog is dominated by movies, with strong contributions from the United States and India. The platform experienced its fastest content growth between 2016 and 2019, and most titles are intended for mature audiences. Drama-related genres remain the most common across the library.